G.3 Comparison of Candidate Trajectory-Prediction Algorithms

The comparison below explains the alternatives in plain language. The highlighted row is the method currently implemented.

Algorithm	How it works	Main advantage	Main limitation	Suitability for this project
Last observation carried forward	Keeps the last known position unchanged.	Almost no computation and works with one record.	A moving aircraft immediately appears stationary.	Poor; useful only as an emergency display fallback.
Constant velocity and heading	Moves the last position using current speed, heading and vertical rate.	Very fast and does not require a training dataset.	Does not smooth noise and cannot adapt well to turns.	Good fallback; it is used when regression cannot be fitted.
Sliding-window linear regression (USED)	Fits straight lines to recent latitude, longitude and altitude values over time.	Simple, explainable, smooths several observations and is inexpensive.	Assumes locally straight motion; accuracy drops during turns or long gaps.	Best present choice for short gaps, local execution and limited data.
Kalman filter	Recursively estimates position and velocity while modelling measurement noise.	Strong noise handling and efficient real-time updates.	Needs a motion model and careful covariance tuning.	Strong next upgrade when receiver-noise statistics are available.
Polynomial regression	Fits a curved time trend instead of a straight line.	Can represent gentle curvature over a short window.	Higher-order extrapolation can become unstable and overfit noise.	Conditional; requires strict order and horizon controls.
Random forest / gradient boosting	Learns nonlinear corrections from engineered flight-state features.	Can model interactions that simple formulas miss.	Requires representative labelled data and does not naturally model sequence continuity.	Research option after a larger verified trajectory dataset is collected.
LSTM / GRU neural network	Learns temporal patterns from complete sequences of prior flights.	Can model complex route and manoeuvre patterns.	Needs substantial training data, tuning and more computing power; less explainable.	Future work for large-scale historical datasets, not the current local baseline.

Table 15: Trajectory Prediction Algorithm Comparison